

Quantifying Circularity in AI Industry

Deal Networks

A Graph-Based Analysis of Investment and Service Flows

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Abstract

The artificial intelligence industry has experienced unprecedented levels of capital growth, with major technology companies participating in complex webs of investments, cloud service commitments, and hardware supply agreements. This paper presents a novel methodology to measure and quantify the flow of circular deals in corporate ecosystems. These are instances where capital or value flows between companies through different mechanisms, creating loops of different lengths. Three complementary metrics are introduced: the *Loop Score*, which measures circularity between company pairs; the *Cycle Score*, which tracks multi-party cycles involving three or more companies; and the *Hub Score*, which aggregates participation across all circular structures to identify systemically central entities. The paper analyzes 91 deals among a curated set of 28 prominent AI, cloud, and semiconductor companies from 2022-2025, and identifies 188 total circular structures consisting of 7 two-party loops and 178 multi-party cycles. The paper's findings show that circular patterns are common among the major AI companies

within the sample, with certain infrastructure providers participating in numerous circular flows. The paper will discuss implications for revenue recognition, valuation interdependence, systemic risk assessment, and market transparency.

Keywords: network analysis, corporate finance, artificial intelligence, circular flows, graph theory, deal networks, hub score, cycle detection, systemic risk, AI Bubble

1. Introduction

The rapid growth of the AI industry has led to huge amounts of capital flowing among multitudes of different technology companies, venture capital firms, and cloud infrastructure providers. Between the years 2022 and 2025, AI companies have received investments upwards of \$100 billion dollars, furthermore, cloud service commitments have reached into the hundreds of billions of dollars (CB Insights, 2024; Bloomberg, 2025).

An important consequence of this capital growth is the emergence of reciprocal relationships between investors and investees. For example, there are cloud providers that invest equity in AI startups, who subsequently commit to purchasing cloud services from the same investors. Hardware manufacturers also invest in cloud infrastructure companies, who then purchase the manufacturer's products. These bidirectional flows create what we call *circular patterns*. These are closed loops where value circulates between parties through different transaction types.

While these circular deals may simply be a result of rational business strategy and vertical integration, they bring up important questions for financial analysts, regulators, and investors. Namely, how should “organic” revenue growth be evaluated when a substantial amount of a company's revenue comes from entities in which it holds equity positions? What are the risk correlations among companies whose valuations depend on contracts with one another?

Research Question

To what extent do circular capital flows characterize the AI industry's corporate deal network, and what are the implications for systemic risk assessment?

This paper offers five contributions. First, it will present a formal methodology for representing corporate deal structures as directed graphs to detect and visualize circular flows. Second, it introduces the **Loop Score**, a composite metric that measures the strength of direct, two-party relationships based on flow type diversity, monetary balance, and data quality. Third, the paper extends this framework to detect and measure multi-party cycles, circular flows involving three or more companies, using depth first search with a complementary **Cycle Score** metric. Fourth, the paper introduces the **Hub Score**, which aggregates participation across both two-party loops and multi-party cycles to identify companies that are central to the industry. Fifth, the paper applies this methodology to a curated dataset of 91 deals among prominent AI, cloud, and

semiconductor companies, to identify circular structures as well as the hub companies that occupy central positions within this network.

2. Background and Related Work

The current AI industry is characterized by a high degree of role overlap among its largest participants. Cloud infrastructure providers also act as investors, compute suppliers, and customers of the companies they invest in. This structural feature, combined with rapid capital growth and increasing market concentration (Stanford HAI, 2024; Vipra & Korinek, 2023), creates conditions where circular capital flows may emerge at scale. Understanding these patterns and their broader implications requires drawing on three bodies of literature: network analysis in financial systems, related-party transaction analysis, and venture capital and platform economics.

2.1 Network Analysis in Finance

Graph-based methods have been widely used within financial models. Allen & Gale (2000) created the foundational model of financial contagion, showcasing how shocks can propagate through interbank lending networks via direct balance sheet linkages. This is a mechanism that the paper then adapts to analyze how valuation shocks might cascade and propagate through investor-investee relationships in AI deal networks. Gai & Kapadia (2010) extended this framework to showcase how highly interconnected nodes can create systemic fragility. The Hub Score method, which quantifies the extent of a company's participation in circular flows, is based on their concept of interconnected

institutions. Acemoglu et al. (2012) demonstrated that shocks to central firms can propagate into aggregate fluctuations when network connections are asymmetric, a dynamic this paper examines in the context of AI deal networks.

2.2 Circular Transactions and Related-Party Analysis

Previous financial literature has studied related-party transactions and how they can affect financial statement quality. Gordon et al. (2004) documented that related party transactions are associated with weaker governance and lower earnings quality; this is a framework directly applicable to the Microsoft-OpenAI relationship, where Microsoft holds equity and receives cloud revenue from OpenAI, where it holds a significant equity position. Dechow et al. (2011) identified financial statement patterns predictive of material misstatements, including unusual revenue growth from related parties. While this paper does not assess the legitimacy of any specific transaction, the Loop Score draws on a similar analytical intuition: that bidirectional flows between related parties warrant closer examination. Circular trading patterns have been studied in equity markets for manipulation detection (Aitken et al., 2015). The paper extends this literature by providing a quantitative framework for detecting circular patterns across heterogeneous transaction types (equity, services, hardware) rather than within a single asset class.

2.3 Venture Capital and Platform Networks

Venture capital literature has documented extensive syndication networks. Hochberg, Ljungqvist & Lu (2007) demonstrated that VC network centrality predicts fund

performance, establishing that investor relationships have real economic consequences beyond capital provision. The AI industry exhibits an intensified version of this phenomenon: cloud providers are not merely investors but also customers and infrastructure suppliers to the companies within their portfolio. This creates multilayer dependencies that are far more concentrated than those usually seen in traditional VC syndication.

Platform economics offers additional theoretical grounding. Rochet & Tirole (2003, 2006) formalized two-sided market dynamics where platforms subsidize one side to capture value from another. Cloud providers investing in AI startups who then consume cloud services exemplifies this logic: equity investments may also have the added function of customer acquisition costs, with returns captured through service revenue rather than traditional exit multiples.

3. Methodology

3.1 Data Collection

To collect the needed data, a curated dataset of 90 deals spanning across 28 companies was constructed from publicly available sources including SEC filings, official press releases and financial news reporting from sources such as Bloomberg, Reuters, CNBC, and The Information. The data collected spanned from January 2022 to January 2025.

The sample companies were selected based on three criteria:

1. Market leadership or valuation exceeding \$1 billion
2. Documented participation within AI related deals from 2022 - 2025
3. Publicly verifiable information via SEC filings or press releases

This dataset was purposefully selected and prominent participants within the industry were used. It is not a random sample of the AI ecosystem. These findings should be interpreted as patterns among the specific dataset and not a comprehensive population estimate.

Each deal record includes the following attributes:

Deal Type (τ): Investment, Cloud Commitment, Supply Agreement, Partnership, Acquisition

Flow Type (ϕ): Money, Compute/Hardware, Service, or Equity

Direction: For each party, whether value flows out (source) or in (recipient)

Amount: Transaction value in USD, where disclosed

Data Status: Confirmed (official announcement with specific figures) or Estimated (reported but unconfirmed)

Confidence Score (c): 1–5 rating based on source reliability and corroboration

Table 1: Dataset Composition by Deal Type:

Deal Type	Count
Cloud Commitment	25

Investment	23
Partnership	24
Supply Agreement	16
Acquisition	3

3.2 Graph Representation

The paper models the deal network as a directed multigraph $G = (V, E)$, where V represents companies and E represents directed edges from deals. Each edge $e \in E$ is characterized by:

Source node (from): The company from which value flows

Target node (to): The company receiving value

Edge type (τ): The deal type classification

Flow type (ϕ): The type of value being transferred

Weight (w): The transaction amount in USD, where disclosed. When transaction amounts are undisclosed, w is set to null. Such edges are retained in the graph for structural analysis but excluded from monetary balance calculations.

Confidence (c): The mean of all source-level confidence ratings across deals contributing to the edge.

When multiple deals share the same source and target companies, they are aggregated into a single edge. Deals involving multiple parties in the same role (e.g., three co-investors in one round) are represented as separate edges. Direction is inferred from

party roles: INVESTOR - INVESTEE, CUSTOMER - SUPPLIER, ACQUIRER - TARGET. Partnership deals without clear directionality are represented as a single edge with a non-directional flag; these edges do not generate reverse edges and therefore cannot independently produce loops.